

NLP

VIVA QUESTIONS

? **What is tokenization? Why is it needed?**

Tokenization is the process of breaking text into smaller units called tokens (words/sentences).
It is needed because machine learning models text directly understand nahi karte, so we convert text into manageable pieces

? Difference between all tokenizers you used?.

Tokenizer	Key Point	Example Input	Output Tokens
Whitespace	Splits on spaces only	"I love NLP"	["I", "love", "NLP"]
Punctuation	Separates punctuation also	"Hello, world!"	["Hello", ",", "world", "!"]
Treebank	Linguistically correct (handles contractions)	"I don't know"	["I", "do", "n't", "know"]
Tweet	Handles emojis, hashtags, mentions	"@user 😊 #fun"	["@user", "😊", "#fun"]
MWE	Combines multi-word expressions	"New York is big"	["New_York", "is", "big"]

? **Why TweetTokenizer is needed?**

Because normal tokenizers fail on emojis, hashtags, and mentions like @user, #AI.

? What is MWE and why important?

Multi-word expressions combine words like "New York" into one token, preserving meaning.

? Difference between Porter and Snowball Stemmer?

- Porter → older, rule-based
- Snowball → improved, faster, supports multiple languages

? What are drawbacks of stemming?

- May produce non-meaningful words
- Example: "studies" → "studi"

? Why lemmatization is better?

Because it uses vocabulary + grammar → gives meaningful base word.

Example: "better" → "good"

? Which lemmatizer did you use?

NLTK WordNet Lemmatizer (most common answer)

? What steps are involved in text preprocessing?

1. Lowercasing
2. Removing punctuation
3. Removing stopwords
4. Tokenization
5. Lemmatization

? What are stopwords?

Common words like "is", "the", "and" which don't carry meaningful information.

? Why remove stopwords?

To reduce noise and improve model performance.

? What is label encoding?

Converting categorical labels into numeric values.

Example:

Positive → 1

Negative → 0

? What is Bag of Words?

Represents text as frequency of words ignoring order.

? Limitation of BoW?

- Ignores context
- Ignores word order

? What is normalized count?

Word frequency divided by total number of words.

? What is TF-IDF and why used?

TF-IDF measures importance of a word in a document relative to all documents.

? Difference: TF-IDF vs BoW?

BoW	TF-IDF
Only frequency	Frequency + importance
No weighting	Weighted

? Why TF-IDF is better?

Because it removes bias of frequently occurring common words.

? What is Word2Vec?

A technique that converts words into dense vectors capturing semantic meaning.

? Why Word2Vec over TF-IDF?

- TF-IDF → no context
- Word2Vec → captures meaning

? Types of Word2Vec?

1. CBOW → context → predict word
2. Skip-gram → word → predict context

? Which one is better?

Skip-gram works better for small datasets and rare words.

? What is embedding?

Numerical vector representation of words.

? What is a Transformer?

A deep learning model used in NLP that uses attention mechanism instead of RNN/LSTM.

? Why transformers are better than RNN?

- Parallel processing possible
- Faster training
- Handles long dependencies

? What is attention mechanism?

It allows model to focus on important words in a sentence.

? What is self-attention?

Each word attends to every other word in the sentence to understand context.

? Components of transformer?

1. Encoder
2. Decoder
3. Attention
4. Feedforward network

? What is positional encoding?

Since transformer doesn't use sequence order, positional encoding adds position information.

? What is morphology?

Study of how words are formed from smaller units (morphemes).

? What is morpheme?

Smallest meaningful unit.

Example:

"unhappiness" → un + happy + ness

? What is Add-Delete table?

A method to transform words by adding/removing suffix/prefix.

Example:

play → playing (add “ing”)

playing → play (delete “ing”)

? Types of morphology?

1. Inflectional → changes tense (play → played)
2. Derivational → changes meaning (happy → unhappy)

? Which libraries did you use?

- NLTK → preprocessing
- sklearn → TF-IDF, BoW
- gensim → Word2Vec
- PyTorch → Transformer

? Why preprocessing is important?

Because raw text is noisy and needs cleaning for better model performance.

? Why use real datasets?

To train models on realistic data and improve generalization



thanks for watching

